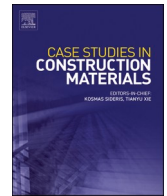




Contents lists available at ScienceDirect

Case Studies in Construction Materials

journal homepage: www.elsevier.com/locate/cscm

Enhancing concrete strength and durability through incorporation of rice husk ash and high recycled aggregate

Mohammad Alharthai^{a,b,*}, Kennedy C. Onyelowe^{c,d,**}, Tariq Ali^e,
Muhammad Zeeshan Qureshi^f, Aïssa Rezzoug^g, Ahmed Deifalla^h, Khalid Alharthi^a

^a Department of Civil Engineering, College of Engineering, Najran University, Najran 66462, Saudi Arabia

^b Science and Engineering Research Center, Najran University, Najran 66462, Saudi Arabia

^c Department of Civil Engineering, School of Engineering and Applied Sciences, Kampala International University, Kampala, Uganda

^d Department of Civil Engineering, Michael Okpara University of Agriculture, Umudike, Abia State 440109, Nigeria

^e Department of Civil Engineering, Swedish College of Engineering and Technology, Wah 47080, Pakistan

^f Department of Civil Engineering, University of Engineering and Technology, Taxila, Pakistan

^g College of Engineering, Imam Mohammad Ibn Saud Islamic University (IMSIU), Riyadh, Saudi Arabia

^h Faculty of Engineering and Technology, Future University, Egypt

ARTICLE INFO

Keywords:

Rice husk ash (RHA)

Recycled concrete aggregate (RCA)

Mechanical characteristics

ABSTRACT

The global environment faces significant challenges due to the massive use of concrete, a dominant material in construction. Addressing this challenge, the present study explores the potential of recycled concrete aggregate (RCA) combined with rice husk ash (RHA) to enhance the durability and strength of recycled aggregate concrete (RAC), particularly in mixes with high RCA content. The research examines RHA as a partial substitute for Ordinary Portland Cement at 5 %, 10 %, 15 % and 20 % replacement levels, alongside RCA substitution rates of 80 %, 90 %, and 100 %. Five RAC mixes were evaluated: Mix-1 served as the control group with varying RCA percentages (80 %, 90 %, and 100 %) without RHA, whereas Mix-2, Mix-3, Mix-4 and Mix-5 incorporated 5 %, 10 %, 15 % and 20 % RHA as a cement substitute. The study evaluated parameters such as workability, compressive strength, tensile strength, water absorption, and acid resistance, in addition to employing non-destructive testing techniques like ultrasonic pulse velocity and rebound hammer tests. After 28 days of curing, Mix-4 (comprising 15 % RHA and 80 % RCA) exhibited the highest compressive strength among the tested RAC mixes, achieving a value of approximately 27 MPa. However, economic analysis revealed that Mix-5 (20 % RHA and 100 % RCA) offers a 31 % reduction in production costs, making it a viable choice for concrete manufacturing. This integration of RCA and RHA presents a sustainable, eco-friendly, and cost-effective approach to enhancing concrete strength and durability.

* Corresponding author at: Department of Civil Engineering, College of Engineering, Najran University, Najran 66462, Saudi Arabia.

** Corresponding author at: Department of Civil Engineering, School of Engineering and Applied Sciences, Kampala International University, Kampala, Uganda.

E-mail addresses: maalharthai@nu.edu.sa (M. Alharthai), kennedychibuzor@kiu.ac.ug (K.C. Onyelowe), bridges.28751@gmail.com (T. Ali), qureshizeeshan746@gmail.com (M.Z. Qureshi), arezzoug@imamu.edu.sa (A. Rezzoug), Ahmed.deifalla@fue.edu.eg (A. Deifalla), eng3333.m.a@gmail.com (K. Alharthi).

<https://doi.org/10.1016/j.cscm.2024.e04152>

Received 26 October 2024; Received in revised form 16 December 2024; Accepted 22 December 2024

Available online 24 December 2024

2214-5095/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

Different social factors have influenced the increased production of waste, such as development of buildings and structures to deplete NA (natural aggregates) and elevated rate in urbanization as well as natural disasters which increased Construction and Demolition Waste (CDW). Disposal of this waste through conventional landfills is not only expensive but also has negative impacts on the environment. The utilization of waste concrete is also very important as it serves as an environmental-friendly measure by preventing the depletion of natural aggregates. Recycled aggregate (RA), which is prepared by crushing waste concrete, can be utilized as a sustainable option to natural aggregates used in construction in partial or full capacity [1–4].

The lower strength and higher porosity of RA (recycled aggregate), which is poor compared to crushed limestone [5–9], lead to low quality in concrete, which includes poor strength and durability of RAC (recycled aggregate concrete). On the other hand, recycled concrete can help reduce the growing shortage of natural concrete aggregate (NCA) while improving the environment, which is helpful for sustainability in the building sector [4,10]. Lima et al. [11] observed that raising the proportion of RA (recycled aggregate) in the concrete mixture led to a decline in compressive strength, and tensile strength. Katz [12] similarly stated that concrete created with NA (natural aggregate) exhibited greater strength than concrete composed entirely of recycled material at the same water-to-cement ratio, leading to decreased compressive strength, and tensile strength. The attached mortar with RA instead of conventional aggregates develops a weak and permeable interface transition zone (ITZ). The ITZs are concrete's weakest points, reducing cement matrix-aggregate bonding. This negatively impacts several attributes of RCA, such as structural performance, strength, and durability [13–18].

Several modification techniques have been investigated to improve the properties of recycled aggregate concrete (RAC) such as using supplementary cementitious materials (SCMs)[19], waste glass powder[20], microbial- and enzyme-induced carbonate precipitation (MICP and EICP)[21]. SCMs enhance the RAC by increasing the density of the concrete and fill the pores, resulting in improved mechanical properties and durability as well as increased cost efficiency; however, they necessitate a closely controlled mix proportion and curing regime. Waste glass powder plays another role of corrosion resistance by virtue of its pozzolan although the effect depends on particle size and which level the glass is incorporated[20]. The MICP and EICP treatments improve RAC by promoting carbonate precipitation to restore pores, increasing the strength and anti-corrosion characteristics, while the process depends on conditions that support bacterial or enzyme action[21]. For this reason, Rice Husk Ash (RHA) is a relatively cheaper and Eco-friendly solution, which enhances RAC through a reaction with calcium hydroxide to form Calcium Silicate Hydrate (C-S-H), a denser and stronger phase. However, while SCMs and MICP/EICP are not easily available and might be difficult to integrate into constructions, RHA is easily available and can be blended easily into constructions and thus, the best solution towards improving the RAC properties environmentally.

The most cost-effective and eco-friendly method to enhance the performance of RCA concrete is by enhancing ITZ between the newly mortar and recycled aggregate using by-products of agriculture and industries. Pozzolanic material is extensively utilized to partially substitute regular OPC in concrete, enhancing its quality and durability characteristics. Using industrial by-products such as granulated blast furnace slag (GGBFS), fly ash (FA), and silica fume (SF) can enhance the durability and mechanical characteristics of RCA [22–25]. However, these materials are encountering shortages because of geographical and technological constraints.

RHA, a byproduct of rice husk combustion, finely ground state, has been extensively utilized as a binder in concrete [26,27]. The utilization of RHA into the production of concrete reduces the amount of cement employed leading to less energy consumption and greenhouse emissions in the production of concrete. The production of cement is a major source of global CO₂ emissions contributing about 7–8 % of the total emissions attributed to the energy intensive calcination process which emits about 0.9 tons of CO₂ per ton of cement [28,29]. However, one of the common materials used in the production of cement, rice husk ash (RHA) which is generated from burning rice husk has a small carbon footprint. It must be noted that when rice husk is burnt, it does emit CO₂, but the amount is very little in comparison to the cement industry. The maximum CO₂ emission of RHA was observed from the data presented by Quispe et al. [30] which showed that one metric ton of RH (rice husk) produce 31.4 kg of CO₂-eq emissions during the agricultural and transportation stage, while the combustion stage emits 4.29 g CO₂-eq. According to Pode [31], the conversion rate of rice husk to ash was around 20 %. The amount of heat produced from the combustion of rice husks was further compared to data published in the literature and was also found to be within acceptable ranges [32].

The partial replacement of cement with RHA in concrete production is a more sustainable approach towards addressing the overall emission problems and environmental impacts of concrete production [33,34]. This makes RHA a viable material for enhancing sustainability for concrete, it not only replaces cement but also a way to utilize agricultural waste.

The properties of RHA like large specific surface area, rich in silica, low cost, internal curing capability and excellent pozzolanic activity, is desirable additional material to improve the mechanical property of RCA [35–40]. Research shows that concrete with RHA integrated exhibited improved compressive strengths in the first 3–7 days as well as after 28 days. [41]. According to Chatveera and Lotwattananuruk [42], using RHA in concrete increases its durability due to which RHA based concrete perform better when exposed to sulfuric acid as well as hydrochloric acid environment. The application of 20 % RHA increased the reaction of hydration at the later phase. It improved the workability, strength and mechanical characteristics of RHA based concrete compared to normal concrete [43, 44]. RHA improves the strength and permeability of RCA by enhancing the microstructure compaction of the concrete matrix [45]. Furthermore, the incorporation of RHA as a partial cement substitute enhances the mechanical properties and durability of concrete while simultaneously lowering the cost of raw materials [46–49].

The integrated blend of Recycled Aggregate (RA) and Rice Husk Ash (RHA) in the production of concrete has been investigated by researchers in past. It has been found that adding 20 % RHA improves the later age hydration reaction in Recycled Aggregate Concrete (RAC) which is beneficial to the mechanical strength and durability of RHA concrete compared to ordinary concretes without RHA.

This is because RHA is a pozzolanic material can combine with calcium hydroxide formed during cement hydration to produce further cementitious materials resulting in increased strength [50]. Similarly the incorporation of 100 % recycled coarse aggregate with 10–15 % RHA has also been found to be suitable for engineering application based upon their satisfactory mechanical performance [51]. Furthermore, it is also reported that the presence of 10 % and 20 % RHA in the concrete mix, results into improved compressive strength compared to ordinary concrete, which has been attributed to the appropriate particle size distribution and the increased density of the concrete matrix [52].

Similarly in RA concrete the additional calcium silicate hydrate (CSH) results from presence of RHA improves the interfacial transition zone (ITZ) and pore structure, thus minimizing pores and achieving a denser structure through pore distribution [53]. The performance of recycled aggregate concrete (RAC) with a high content of RCA and the incorporation of RHA needs to be studied comprehensively. This research focuses on assessing fresh properties, mechanical behaviour, and non-destructive testing methods of the RAC. Additionally, the study aims to explore the potential environmental and economic benefits of using RHA as a filler and cement substitute in RAC.

2. Novelty of the study

Unveiling the Potential of RHA in RAC Early studies investigating recycled aggregate concrete often employed random replacement levels of recycled aggregates or varied percentages at specific intervals; while these approaches provided valuable insights, they may need to fully capture the impact of higher RCA content on concrete properties. This is particularly critical as the construction industry strives towards increased utilization of recycled materials. This research fulfils the gap in literature by examining the effects of integrating high RCA (80 %, 90 %, and 100 %) along with varying proportions of RHA (5 %, 10 %, 15 %, and 20 %) in concrete. Previous studies have explored RCA or RHA independently; however, comparative research on the combined impact of both ratios at high replacement levels is scarce and lacks evidence on mechanical strength, sustainability, and cost analysis. The work may also be useful because it not only outlines technical analysis of the materials but also makes an additional focus on cost-effectiveness of the concrete production supplemented with a cost-benefit analysis.

Pakistani contribute around 1.2 million tonnes per annum to the global rice husk production while global production was estimated to be ranging between 130 and 170 million tons during 2020–21. Investigating the application of using rice husk particularly in conjunction with Recycled Concrete Aggregate (RCA) provides great benefits to sustainable construction. This current research therefore uses a new statistically based methodology of evaluating the changes in strength of concrete with incorporation of both rice husk ash (RHA) and high RCA content concrete. Rice husk silica with increased pozzolanic activity improves the chemical structure of concrete's mechanical properties and Reclaimed construction waste use contributes to sustainability in this concrete mixture. This integration not only minimizes environmental effects but also enhances the concrete quality and durability with a tough option to the regular materials. This research aims to determine the most effective amounts of RHA and appropriate combinations of RAC that can guarantee excellent performance along with enabling the formulation of low cost sustainable concrete, which is suitable for structural application.

3. Methodology

3.1. Materials

3.1.1. Binders

This study used the 'Fauji' Cement brand Pakistan, which meets the ASTM C-150 standards [54]. RHA was purchased from Sika Ltd Pakistan. The ash was manually cleaned in the lab and purified using sieve #4 and filter #40. The material achieved the appropriate fineness after further processing in the ball mill. The chemical composition is displayed in Table 1.

3.2. Aggregates

The fine aggregate used in this study was sourced from Lawrencepur, Pakistan. The fineness modulus and specific gravity (ASTMC128–79 [55]) were determined to be 1.66 and 2.71. The sand absorbed 1.20 % of water during a 24-hour immersion period. Margalla crush of high quality was utilized in this study. The specific gravity was 2.68 according to ASTM C 127–81 [56]. Over 24 hours, coarse aggregate absorbed 0.80 % of water. Coarse aggregate has a moisture content of 1 % according to ASTM C 566–78

Table 1
RHA and OPC Chemical Composition.

Chemical Composition	RHA	Cement
Al ₂ O ₃	6.72	4.26
SiO ₂	76.44	21.1
Fe ₂ O ₃	4.75	3.48
MgO	1.56	2.47
CaO	3.53	64.15
SO ₃	0.32	2.22
K ₂ O	2.68	1.04
Na ₂ O	1.26	—
LOI	2.74	1.27

[57]. The moisture content of the dry aggregates was below 10 %.

4. Experimental program

To achieve the objective of the study, 12 concrete mixes were prepared by varying the use of RCA at 0 %, 80 %, 90 %, and 100 % by weight replacement of natural coarse aggregate. Moreover, RHA was blended with the Ordinary Portland Cement (OPC), which constituted 10 % and 20 % of the total OPC by weight. The materials and experimental work in this study is illustrated in Fig. 1.

The testing carried out in this investigation is shown in Table 2. Table 3 lists the casting schedule and test specimens while Table 4 summarizes the mixed proportions in concrete batches and their corresponding designations. The details are determined by the weight percentage (wt%) of RCA (0 %, 80 %, 90 %, and 100 %) and RHA (5 %, 10 %, 15 %, and 20 %) used in each batch. Each mix Mix-1, Mix-2, Mix-3, Mix-4, and Mix-5 is characterized by these specific replacement percentages. Each ID varies based on the used mix as follows:

- For Mix 1: 0 % RHA is used
- For Mix 2: 5 % RHA is used
- For Mix 3: 10 % RHA is used
- For Mix 4: 15 % RHA is used
- For Mix 5: 20 % RHA is used

In this study, the high-water absorption rate of RCA was considered to adjust water content during the mix design. RCA has a higher absorption capacity than NCA because they come from crushed concretes, and they contain interlocking among themselves which can limit the availability of free water in the concrete mix affecting the workability and the strength of the concrete. To address this aspect, the water to cement ratio was well regulated; and as needed for the incorporation of RCA, water content required was added to compensate the water that RCA was expected to absorb. Further, the water content in the mix was planned to support enough water for the cement to be capable of doing its hydration with RCA having a greater absorption capacity, considering the ratio appropriate to retain the necessary water content for both cement and RCA in the concrete and at the same time necessary to produce the sufficient workability. This approach ensures that the increased water demand of RCA does not affect the properties of concrete in any way.

5. Results and discussion

5.1. Workability

A slump test was performed to determine the effect of RHA content on the workability of concrete, where the difference in the slump value is evaluated with different percentages of RHA that may be added into different mixes. Fig. 2 demonstrates the trend of

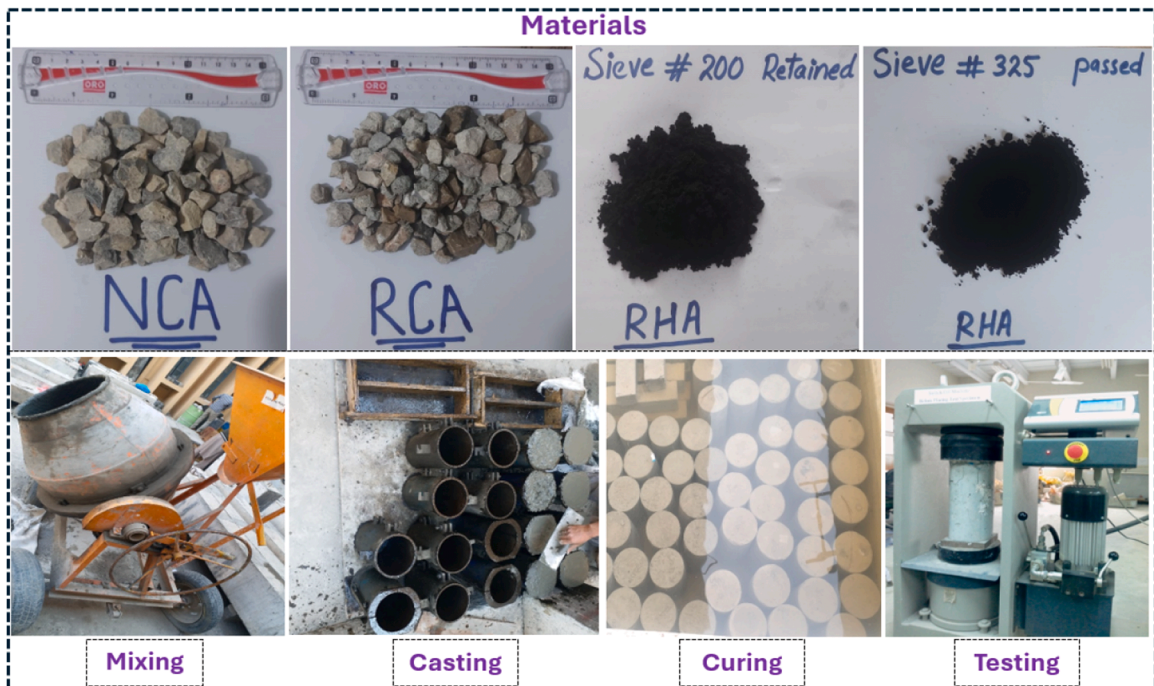


Fig. 1. The material used, and their experimental work in this study.

Table 2

The test performed in this investigation.

Workability	Compressive Strength	Tensile Strength	Water absorption	Acid Attack	Ultrasonic Pulse Velocity	Rebound number
ASTM C143	ASTM C39	ASTM C496	ASTM C642	ASTM C1898	ASTM C597–16	ASTM C805

Table 3

Specimens and Testing Details.

Testing	Detail of Specimens	Curing Periods (days)			Number of Samples	
		7	28	90	Individual (Per Mix)	Total (Per 3 Mixes)
Compressive Strength	150 × 300 mm	3	3	3	9 × 4 = 36	108
Tensile Strength	150 × 300 mm	3	3	3	9 × 4 = 36	108
Water Absorption	150 × 300 mm	2	2	2	6 × 4 = 24	72
Acid Attack	150 × 50 mm	2	2	2	6 × 4 = 24	72
UPV	150 × 300 mm	2	2	2	6 × 4 = 24	72
Rebound Hammer	150 × 300 mm	2	2	2	6 × 4 = 24	72
Grand Total						504

Table 4

Composition of Concrete Blends.

Mix ID	OPC (kg/m ³)	Fine Aggregate (kg/m ³)	NCA (kg/m ³)	RCA (kg/m ³)	RHA (kg/m ³)	Water (kg/m ³)
MX1-RCA0	440	580	1115	0	0	245
MX1-RCA80	440	580	223	892	0	245
MX1-RCA90	440	580	111.5	1003.5	0	245
MX1-RCA100	440	580	0	1115	0	245
MX2-RCA0	418	580	1115	0	22	245
MX2-RCA80	418	580	223	892	22	245
MX2-RCA90	418	580	111.5	1003.5	22	245
MX2-RCA100	418	580	0	1115	22	245
MX3-RCA0	396	580	1115	0	44	245
MX3-RCA80	396	580	223	892	44	245
MX3-RCA90	396	580	111.5	1003.5	44	245
MX3-RCA100	396	580	0	1115	44	245
MX4-RCA0	374	580	1115	0	66	245
MX4-RCA80	374	580	223	892	66	245
MX4-RCA90	374	580	111.5	1003.5	66	245
MX4-RCA100	374	580	0	1115	66	245
MX5-RCA0	352	580	1115	0	88	245
MX5-RCA80	352	580	223	892	88	245
MX5-RCA90	352	580	111.5	1003.5	88	245
MX5-RCA100	352	580	0	1115	88	245

variation. The pattern observed for RHA shows that as the amount of RHA replacement increases, a decrease in workability is observed across the different RAC Mixes (Mix-1, Mix-2, Mix-3, Mix-4 and Mix-5). Mix-1 having no RHA, with varying levels of RA (80 %, 90 and 100 %), exhibits a decrease in a slump value (24.4 %, 44.4 %, 61.1 % respectively) compared to the control mix.

Mix-2, which contains 5 % RHA, showed a reduction of 26.1 %, 48.8 %, and 63.6 % in slump value compared to the control mix. Similarly, Mix-3, containing 10 % RHA, with different levels of RA (80 %, 90 %, and 100 %), showed a decrease in a slump value (26.9 %, 53.8 %, and 67.8 %, respectively) compared to the control mix. Furthermore, Mix-4 and Mix-5, containing 15 % and 20 % RHA, exhibit reductions of 27.7 %, 57.3 %, and 70.5 %, and 25.6 %, 55.7 %, and 67.1 %, respectively. The decrease in performance occurs from the fact that the RCA has an open and random pore structure that reduces the bond strength at the interfacial transition zone, as well as increases water uptake, which slows the cement hydration process. Besides, non-uniform and fine nature of RHA particles resulted in increased water requirement, which influenced workability and compaction characteristics. These factors combined erode concrete performance. A higher RHA content also leads to a greater specific surface area of the binding agents. This means more surface area for the binders to interact with water. As a result, the binders absorb additional water, decreasing the total free water in the concrete structure. This reduces the slump and overall workability [58–60].

5.2. Compressive strength

Fig. 3 shows the compressive strength of 14, 28-, and 90-days using RHA-based concrete. The control sample (MX1-RCA0) has the maximum strength with values of 25.8 MPa, 32.6 MPa, and 35.4 MPa at 14, 28, and 90 days, respectively. The 100 % RCA (MX1-RCA100) has the lowest strength, measuring 12.6 MPa, 16.2 MPa, and 17.3 MPa on the same days of curing. This phenomenon is

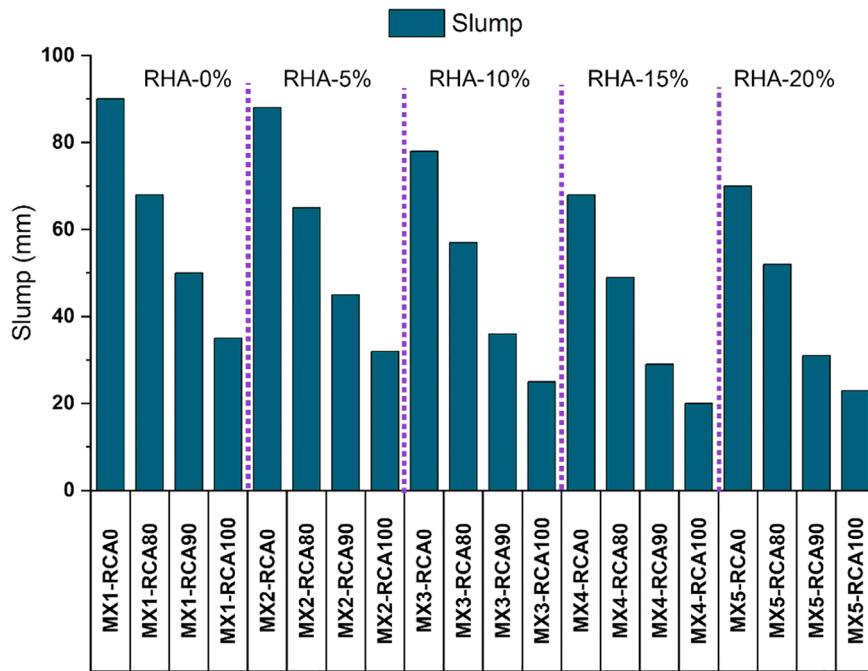


Fig. 2. Results of Workability using different percentages of RHA and RCA.

attributable to the increased porosity of old mortar and cement paste attached to the RA (recycled material) [61,62]. Adding 5 % RHA to the control mix increased the strength by 8.5 %, 9.2 %, and 7 % for the same days of curing. The maximum strength is achieved by including 15 % RHA in a control mix, which is 13.5 %, 15.6 %, and 11.8 %, respectively. Further, the addition of 15 % RHA with 100 % RCA increased the strength of the mix by 44 %, 41 %, and 45.5 % compared to the control mix. This is due to RHA's greater fineness enhanced its interaction with $\text{Ca}(\text{OH})_2$ resulting in more calcium silicate hydrate (C-S-H), leading to a higher compressive strength [63]. The enhancement of strength in recycled concrete aggregate is predominantly attributed to the pozzolanic impact of RHA particles [64]. The pozzolanic reaction between free calcium hydroxide and rice husk ash decreased the porosity of the paste [65]. The

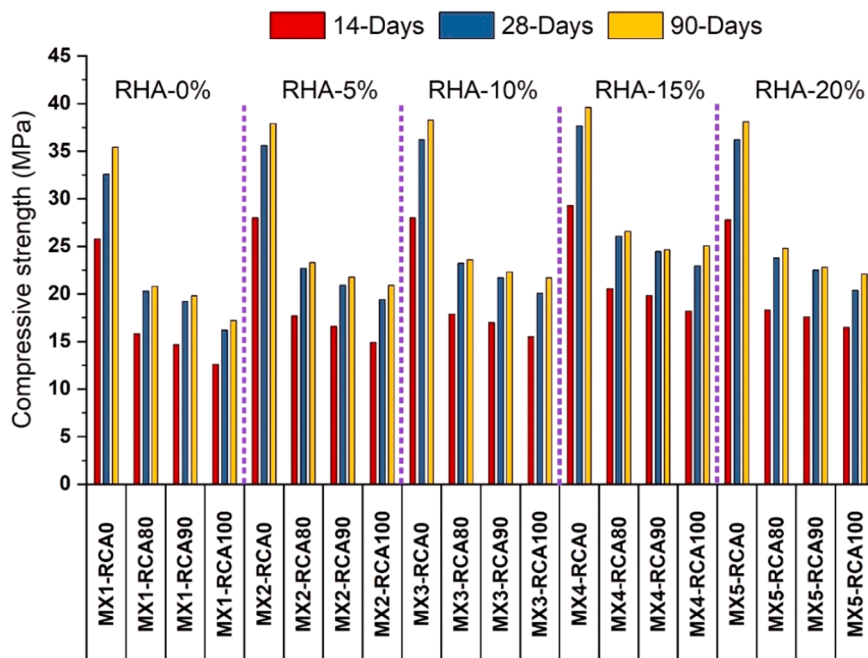


Fig. 3. Compressive Strength after 7,28 and 90-days curing.

porous structure of RA absorbed a significant quantity of water and reduced the matrix's free water content, weakening the pozzolanic impact of the RHA particles and reducing the overall volume of hydrated materials, which slows concrete strength growth during curing. The increase in concrete strength at various curing ages results from the filling effect, cement hydration reaction, and the pozzolanic effect of the particles of RHA [66,67].

Previous literature revealed a similar result, indicating that the mechanical characteristics were most favourable, with an RHA percentage ranging from 10 % to 20 %. The compressive strength begins to decline whenever the quantity exceeds 20 % [68]. Sata et al. [69] observed that the mortar's mechanical characteristics were at their best when the percentage of RHA varied from 10 % to 20 %. The strength of mortar decreases with content levels above 20 %. Ganesan et al. [70] revealed that 20 % RHA concrete has the maximum compressive strength. It has been reported by Kannan et al. [71] that the compressive strength dropped when the ratio of replacement of RHA was considered 20 %.

Details of the compressive strength of concrete mixtures incorporating Rice Husk Ash and Recycled Coarse Aggregate are important in evaluating the performance of the concrete. To establish the correlation between these variables, several concrete samples were casted and tested at the age of 28 days, and the results were graphed, where each point correspond to the average of the three measurements. A linear regression analysis was conducted to analyze the trends and to determine the viability of the predictions made by the model.

5.3. Tensile strength

Fig. 4 displays the findings of tensile strength at 14, 28, and 90 days. The control mix exhibited tensile strengths of 2.18 MPa, 2.87 MPa, and 3.14 MPa after 14 days, 28 days, and 90 days, respectively. The tensile strength of samples using 100 % RCA was measured at 1.27 MPa, 1.97 MPa, and 2.14 MPa after the same number of days of curing. Strength improved by 2 %, 3.9 %, and 6.6 % when 5 % RHA was added to 100 % RCA in comparison to the control mix. With the incorporation of 15 % RHA and 100 % RCA, the strength increased by 17.7 %, 21.3 %, and 22.3 % after curing for 14 days, 28 days, and 90 days, respectively. In previous research, it has been revealed that the tensile strength of concrete determined approximately 8–14 % of the compressive strength [72]. The tensile strength of normal concrete in this investigation was 9.53 % of its compressive strength. The split tensile strength of RAC was found to be like that of concrete prepared from normal aggregates. The compressive strength ratio to tensile strength in recycled aggregate concrete varied between 8.9 % and 12.1 % [73]. As the curing age increases, the particles that are active in RHA react with OPC to ensure strong development. Furthermore, RHA serves an effective filling function in cement, which reduces the hard pores of the cement slurry and enhances its density. This may be one cause for the significant increase in tensile strength. In addition to the strength of the aggregates, the bonding force among them was the key factor allowing the mixture to withstand splitting failure. In addition to the strength of the aggregates, the bonding force among them was the key factor allowing the mixture to withstand splitting failure. In RAC, the interfacial transition zone is rather weak, and is comprised of the cement matrix, the aggregate particles and the interface between them. Common defects occur at the ITZ, where recycled aggregates contain old mortar attached, and micro-cracks exist, thus exhibiting poor bonding and strength. These defects make the ITZ the most critical region for failure of RAC when exposed to

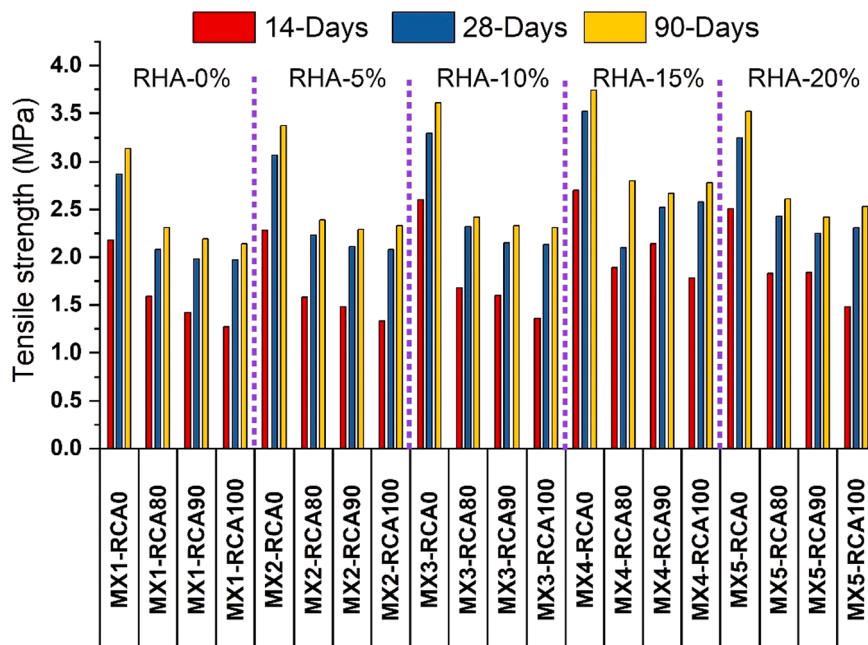


Fig. 4. Tensile Strength after 7,28 and 90-days curing.

mechanical loads. While mixing and moulding the mixture specimen, the recycled aggregate might absorb more water. The micropores on the recycled aggregate surface allow hydration products to be embedded to improve bonding between aggregate particles, increasing the mixture's splitting strength [74]. The pictorial view of RCA used in this study is shown in Fig. 5.

5.4. Water absorption (WA)

WA is a significant test for assessing the durability of concrete as it provides a measurement indirectly of the porosity that is accessible to water [75,76]. The water absorption using an immersion test measures the difference in mass between an SSD specimen and the sample's weight in a dry condition. The water absorption by immersion of concrete mixes is primarily influenced by parameters that regulate the porosity of concrete, specifically the water-to-cement ratio (w/b) [75]. The results of absorption are shown in Fig. 6. Despite the RCA effect, the water absorption through immersion reduced as the RHA incorporation level increased. The results indicate that water absorption decreased from 5.2 % to 4.3 %, 3.9, 3.2 and 3.5 % with the addition of 5 %, 10 %, 15 % and 20 % RHA at 28 days, respectively. Under the same condition, in the case of 100 % RCA the water absorption reduced from 7.6 % to 6.9 %, 6.3 %, 4.9 % and 5.3 % on the same days, respectively. However, it is evident from the absorption graph that there is a decreasing trend in comparison with the curing periods (14, 28 and 90 days) irrespective of the mix type.

For the control mix (MX-RCA0), the absorption at 14, 28, and 90 days is 7.5 %, 5.2 %, and 4.4 %, respectively. For the same mix with the addition of 15 % RHA 5.3 %, 3.2 %, and 2.7 % absorption is calculated on the same days, while with the addition of 20 % RHA, the absorption is 4.7 %, 3.5 %, and 3.3 %, respectively. Utilizing 80 %, 90 %, and 100 % RCA the initial absorption found is 8.9 %, 7.6 %, and 6.7 %; in the case of the 15 % RHA the absorption is 6.6 %, 4.9 %, and 4 %, respectively while in case of 20 % RHA, the absorption is 6.1 %, 5.3 % and 4.6 % after 14, 28 and 90 days of curing respectively. The addition of rice husk ash (RHA) into concrete and the substitution of NA with RA greatly assists in reducing the proportion of water absorption that arises from the porous nature of RA in concrete. Although the highly porous nature of RA causes the concrete to have higher water absorption, the pozzolanic nature of RHA reduces this effect by filling the pores and increasing the density of the concrete structure. It has been observed that replacing about 15–20 % of cement with RHA can reduce water absorption in RA-based concrete and enhance its durability and resistance to water penetration [77,78].

5.5. Acid attack

The results of Acid resistance in terms of compressive strength loss after one month and three months in a 5 % HCL environment are shown in Fig. 7. The findings revealed that the compressive strength after 7 days for the control mix was 22.7 MPa, which was found to reduce to 10.7 % and 31.6 % after one and three months of curing in an acid environment. The compressive strength for 100 % RCA is found to be 8.4 MPa, which reduces by 17.3 % and 39.7 % after the same curing exposure. In Mix-4 (15 % RHA) with 100 % RCA, the 7-day strength is found to be 12.3 MPa, which reduces 12.8 % and 31.1 % after 1 month and three-month curing. Furthermore, for Mix-5 (20 % RHA), the 7-day strength with 100 % RCA is found to be 11.3 MPa, which reduces 12.3 % and 31.2 % after the same periods of assessment. The findings of the acid resistance tests conducted on RHA concrete can be explained in terms of the characteristics of rice husk ash (RHA) and its integration with the cement matrix. Although the property of RHA as a pozzolanic material

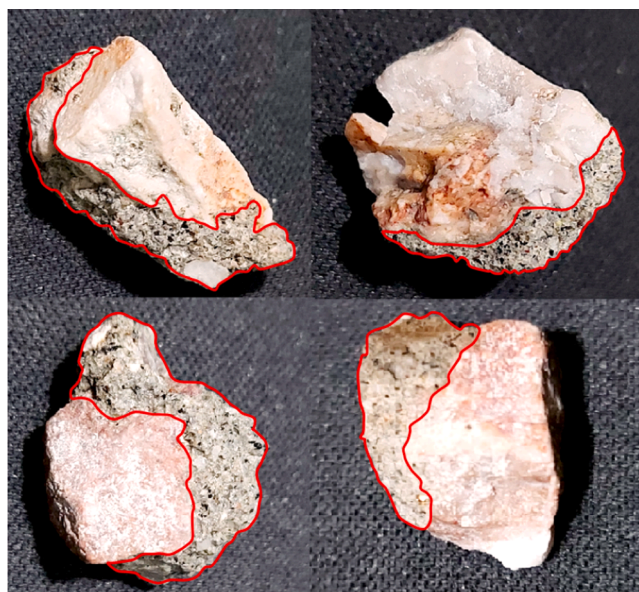


Fig. 5. The pictorial view of using RCA.

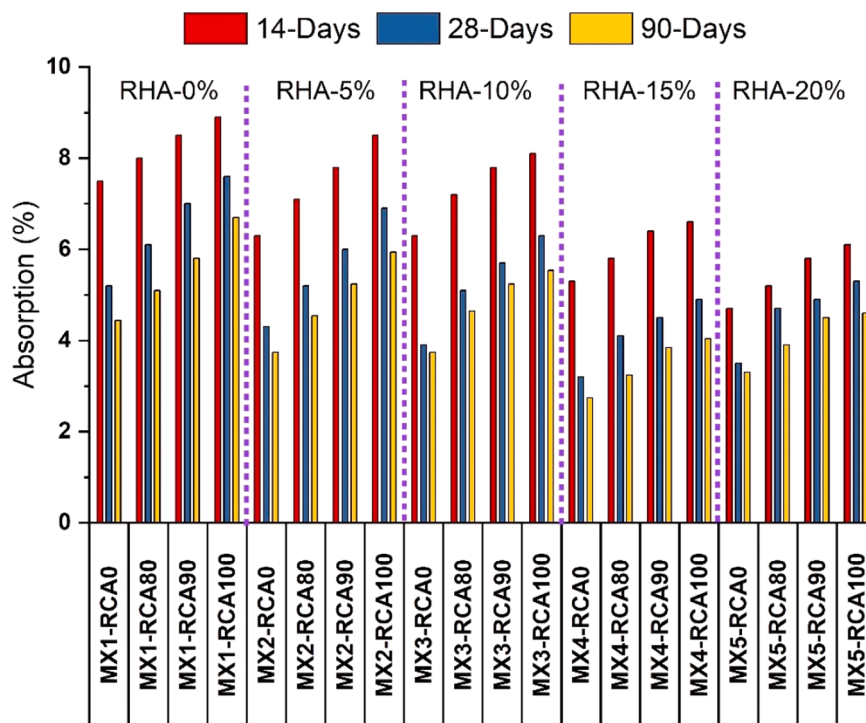


Fig. 6. Results of absorption at 14,28 and 90-days curing.

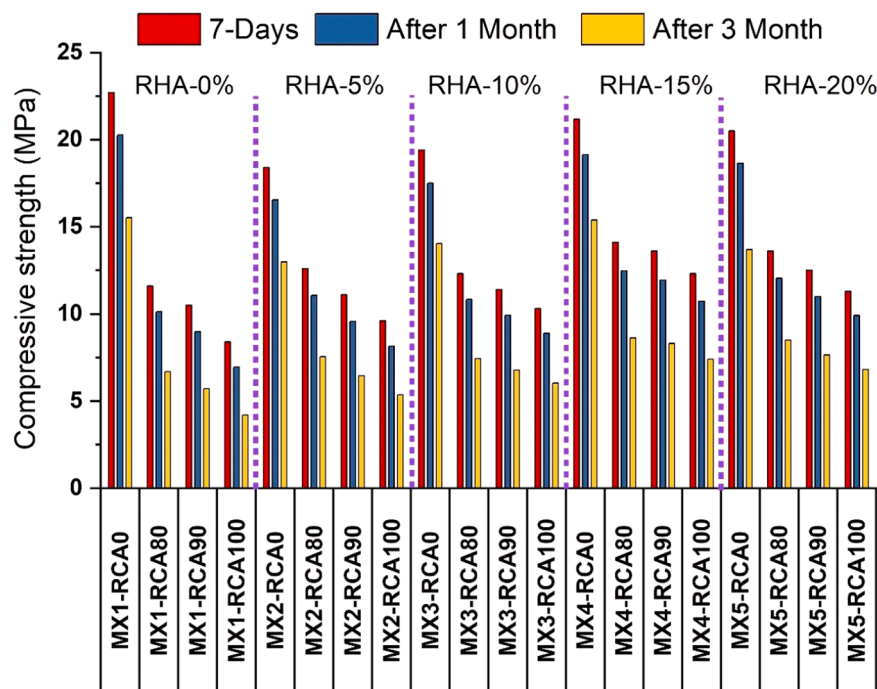


Fig. 7. Results of Acid Attack.

providing concrete strength through formation of C-S-H gel, the increased porosity and reduced density facilitate less resistance to acid attack. Acid attack commonly results in the dissolution of calcium hydroxide ($\text{Ca}(\text{OH})_2$) and strip calcium ions out of the concrete matrix diminishing its mechanical properties [79]. In case of RHA concrete, the pozzolanic reaction between RHA and $\text{Ca}(\text{OH})_2$ limits the amount of free calcium hydroxide in concrete which is very important for the concrete protection against acidic environment [80].

Moreover, the larger pore structure and increased porosity of RHA concrete especially at higher replacement levels (15–20 % RHA concrete) provide more access points for the penetration of acidic solutions which in turn aresses the degradation process [81].

Similar findings revealed that using M30 Grade concrete acid resistance found a weight loss of 1.5 % after 28 days of exposure to 5 % H_2SO_4 [82]. It has been observed that NAC had a 63 % strength loss and RAC a 67 % strength loss when exposed to a 3 % HCl solution. The addition of 30 % rice husk ash decreased these losses to 39 % [83]. Sanjukta Sahoo et al. [84] found an 8 % decrease in strength when exposed to 1 % H_2SO_4 .

5.6. Ultrasonics Pulse Velocity (UPV)

The UPV test is a valuable non-destructive method for assessing the uniformity and compressive properties of concrete structure [85]. Cracks in concrete delay the speed of sound waves. The ultrasonic test is used to measure the speed of sound to evaluate the quality of concrete. A lower speed suggests the presence of potential faults. [86].

Fig. 8 shows the findings of UPV. This research shows that the UPV decreases for all RCA percentages (80 %, 90 %, and 100 %) regardless of the concrete age. However, the results of Mix-4 and Mix-5 demonstrate that utilizing RHA can improve the value of the concrete containing RCA. The UPV value for the control sample in Mix-1 after 28 days is 4141 m/sec. This value decreases to 3474 m/sec for 80 % RCA, 3196 m/sec for 90 % RCA, and 2818 m/sec for 100 % RCA. Similarly, the Mix-4 with 15 % RHA and no RCA exhibited a UPV value of 4346 m/sec. This value decreased to 3885 m/sec, 3559 m/sec, and 3206 m/sec with 80 %, 90 %, and 100 % RCA, respectively. In the same way, the UPV value for Mix-5 with 20 % RHA and no RCA was 4214 m/sec. However, it dropped to 3772 m/sec, 3456 m/sec, and 3095 m/sec with 80 %, 90 %, and 100 % RCA, respectively. The UPV value for the control sample after 90 days in Mix-1 is 4606 m/sec, which drops to 3933 m/sec for 80 % RCA, 3699 m/sec for 90 % RCA, and 2967 m/sec for 100 % RCA. Additionally, the UPV value for Mix-4 with 15 % RHA and no RCA was 4775 m/sec, reduced to 4327 m/sec, 3918 m/sec, and 3291 m/sec with 80 %, 90 %, and 100 % RCA, respectively. Similarly, Mix-5 with 20 % RHA having no RCA showed a UPV value of 4728 m/sec which reduced to 4293 m/sec, 3896 m/sec, 3261 m/sec for 80 %, 90 %, and 100 % RCA, respectively.

Adding RHA to the cementitious matrix and RCA introduces micro-fillers. its fine particles effectively fill micro-pores. This pore-filling process creates a more compact and cohesive structure, leading to improved UPV measurements [87].

5.7. Rebound Number (RN)

The findings of the RN test for all concrete mixes after 28 days, are given in Fig. 9. The analysis indicates that the RN of concrete blends reduces as the ratio of recycled concrete aggregate increases. The RN of control concrete in Mix-1 is 30 and 36 after 28 and 90 days, respectively. This value decreases to 25 and 28 for concrete having 80 % RCA while for 90 % RCA, the RN reduces to 24 and 32; furthermore, for 100 % RCA, the value of RN dropped to 21 and 27 after the same curing days.

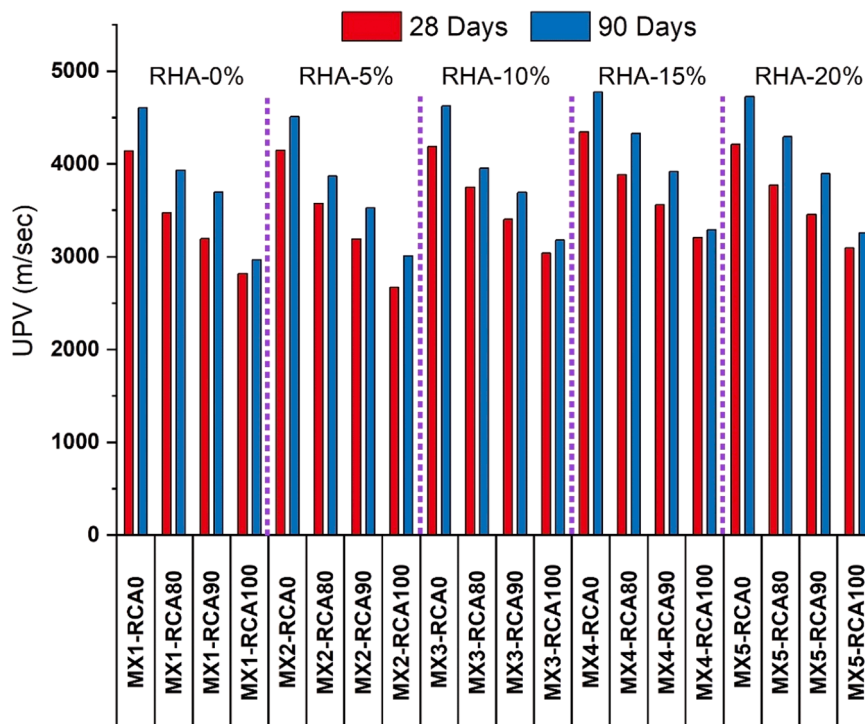


Fig. 8. Results of UPV after 28 and 90days.

Similarly, the RN of Mix-4 having 15 % RHA is 31 and 35 for control concrete after 28 and 90 days respectively. The RN decreases to 27 and 32 for concrete containing 80 % RCA, while for 90 % RCA; the RN reduces to 24 and 29; furthermore, for 100 % RCA, the value of RN dropped to 21 and 24 after the same days of curing. In addition, the RN of Mix-5, which contains 20 % RHA (Rice Husk Ash), is 30 and 34 for the control concrete after 28 days and 90 days, respectively. The RN decreases to 28 and 31 for concrete having 80 % RCA. For concrete containing 90 % RCA, the RN decreases to 25 and 29.

The RN of the concrete mix is influenced by the density, surface hardness, and compressive strength of the concrete. The inclusion of more porous and weaker RCA in the concrete leads to a reduction in these qualities, causing a fall in the rebound number. There is a consistent pattern that shows the qualities and content of the RCA affect the rebound number value [88,89].

Using RCA in concrete led to a reduction in Schmidt Hammer rebound values because of a combination of variables [90–92]. The ease of compacting RCA and the reactions it undergoes affect concrete stiffness, influencing the return of elastic energy more than with new aggregates [92]. Additionally, the uneven surface roughness of RCA can impact the contact area and the transfer of energy during the test, thereby causing a decrease in results [93]. Furthermore, the increased hygroscopicity of RCA in comparison to unused aggregates might result in the retention of moisture, which acts as a damping agent and absorbs impact energy, hence further decreasing the rebound number [92,94,95]. However, a smaller rebound number would not necessarily imply weaker RCA concrete. Analyzing results may require calibration modifications, and the rebound number can be considerably affected by the individual features of RCA (such as source material and processing), as well as the total mix layout and curing circumstances [92]. Thus, it is essential to conduct a thorough examination that considers these previously mentioned elements to effectively determine the quality of RCA concrete employing the Schmidt Hammer testing [92].

6. Microstructure of RCA -Rice Husk Concrete

The properties of Recycled Aggregate Concrete (RAC) are generally lower bond strength in the Interfacial Transition Zone (ITZ) and a higher porosity of the microstructure than normal concrete. These deficiencies reduce mechanical strength and durability of the concrete. However, the incorporation of Rice Husk Ash (RHA) has been observed to minimize these problems with enhanced microstructure of ITZ and insurgence of porosity. In the composition of concrete as RHA is pozzolanic in nature, it plays a very vital role in increasing bond strength and diminishing the permeability of the concrete thus making the concrete matrix denser and more durable in nature.

The study reveals that microstructure of RAC prepared with different replacements of RHA is shown in Fig. 10. For the control sample (RA2) has a matrix with many pores, and large rod-shaped calcium hydroxide (CH) crystals. In turn, the sample with 10 % RHA (RA2 10) exhibit higher density, less porosity and the absence of the CH crystals. This improvement of the microstructure can be explained by the pozzolanic characteristic of RHA. Because the silica content is high and specific surface area is large, RHA offers numerous nucleation sites for cement hydration to consume CH during the pozzolanic reaction. This process creates more of the calcium silicate hydrate (CSH) gel, which helps to further alter the pore size distribution and to enhance the interface between

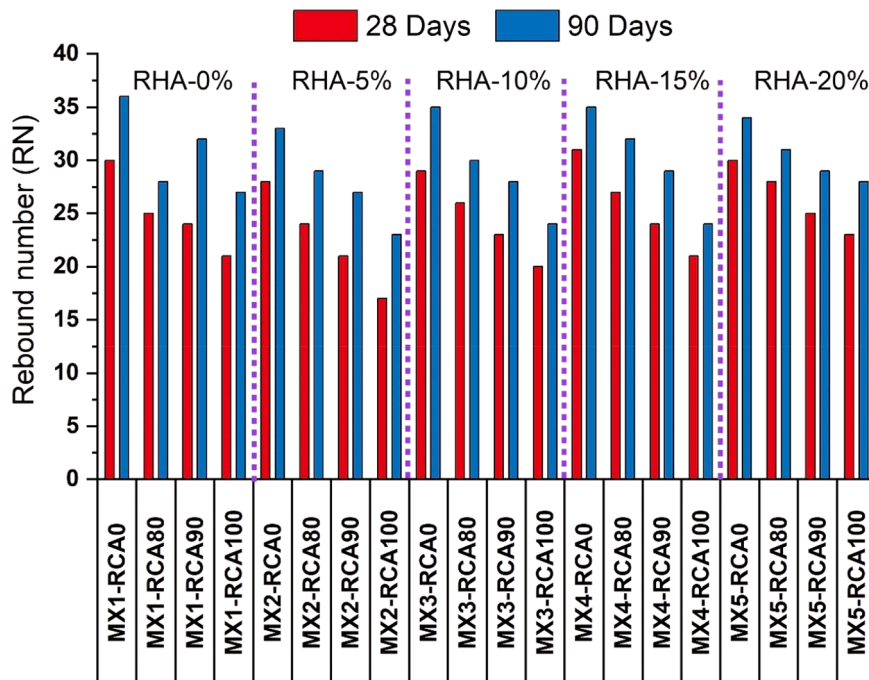


Fig. 9. Results of Rebound Hammer after 28 and 90 days.

cementitious matrix and recycled aggregate [58]. This enhancement in the microstructure corroborates with other studies in the literature where RHA has been identified to cure the issue of porosity, density the cement matrix and enhance the ITZ of concrete [96, 97].

In the case of the RA2 20, which contains 20 % RHA, the microstructure revealed a massive compact region within the matrix; however, discrete pore formation has been observed in the periphery. Higher RHA content causes an even greater surface area (SA), which contributes to a greater adsorption of water, but at the same time, prevents the further hydration and pore filling in these areas. Also, several internal pores are seen in the matrix, which may be caused by the honeycomb like surface appearance of the larger RHA particles [58]. This agrees with the references [98,99], where it was stated that higher than optimal amounts of RHA can swell and cause difficulties when it comes to cement hydration and pore size reduction while at the same time improving the microstructure of the concrete [98,99].

Literature references show that incorporating RHA in concrete not only enhances mechanical properties but also increases durability through structural regulation. Research has shown that incorporating RHA results in reduced permeability and increased resistance against chloride ions and other active agents due to changes in the pore structure [100,101]. Therefore, the incorporation of RHA beneficially increases not only the RAC density but also its long-term performance and durability.

Fig. 11 reveals that 0 % RHA paste having high number of rod-shaped CH crystals and many pores which lead to formation of a loose surface structure. This is probably because of poor interaction of cement particles with water resulting in weak packing density as supported by earlier literature [102,103]. The paste containing 10 % RHA has a more compact structure with no indications of pores, however, some cracks have been identified in the interfacial transition zone. The 20 % RHA paste is equally workable and denser with only negligible micro cracks present at the interface between the aggregate and mortar, and a limited presence of CH crystals. The smaller particle size of RHA tends to increase the specific surface area of the material and hence leading to more cement hydration nucleation sites. This leads to the creation of more C-S-H gel and an enhancement of density and strength of the concrete matrix. This concept is in alignment with studies carried out on pozzolanic materials which underscore their role in terms of improving the inherent workability and microstructure of concrete. The reaction between CH resulting from hydration of the binder and the amorphous silica in RHA leads to the best packing of C-S-H gels, better bond in the interfacial transition zone, less crack formation and hence better mechanical properties in the resulting concrete [104,105].

7. Cost-benefits of RCA -Rice Husk concrete

This paper demonstrates the viability of using Rice Husk Ash (RHA) and Recycled Coarse Aggregate (RCA) in concrete prices by analyzing their cost benefit over conventional concrete which utilizes cement and natural coarse aggregate (NCA) per cubic meter of concrete. The cost analysis of each blend are showed in Table 5. Specs for the baseline mix, MX1-RCA0 was 440 kg cement and 1115 kg NCA per cubic meter it has total cost of PKR 31040. It is the scale for assessing the cost effectiveness of the other mixes which comprise different ratios of RHA and RCA.

When RHA is incorporated with cement partially, the total cost of concrete comes down when RHA is used as a partial replacement for cement. For instance, in MX2-RCA0, 22 kg of RHA replaces some cement, thus reducing cement by 418 kg from the original mixture, and the total cost by 30,556 PKR, giving approximately 1.56 % cost benefit. The cost reductions increase proportionately as the portion of RHA rises. When the quantity of RHA is 44 kg in MX3-RCA0, the total cost becomes 30,072 PKR and there is 3.12 % cost saving. In the same manner, MX4-RCA0 and MX5-RCA0 utilize 66 kg and 88 kg of RHA, thereby lowering the total cost to 29588 PKR and 29104 PKR which is 4.68 % and 6.24 % cheaper in cost respectively. These savings are obtained only by incorporating a part of the cement with cheap material known as RHA.

This enables even greater cost cuts when RCA is used instead of NCA, in MX1-RCA100, where RCA fully replaces NCA, the total cost drop is 23,235 PKR, which indicates 25.14 % cost saving compared to the baseline. The reduction proves that RCA is cheaper as a resource compared to the natural aggregates that it replaces. The cost saving is further enhanced and realized when both RCA and RHA are applied simultaneously. MX2-RCA100 with 22 kg of RHA added having only 1115 kg RCA cuts the total cost to 22751 PKR, a cost advantage standing at 26.70 %. The savings increase as a ratio to RHA rises even further. MX3-RCA100, incorporating 44 kg of RHA, costs 22,267 PKR with a cost benefit of 28.26 %; on the other hand, MX4-RCA100 with 66 kg of RHA costs only 21,783 PKR with the cost benefit of 29.82 %. The highest cost benefit is achieved in MX5-RCA100 where 88 kg of RHA is mixed with 100 % RCA, the total cost of which is 21,299 PKR with an overall cost benefit of 31.38 %.

From the results, there are large savings in the cost of concrete when RHA and RCA are blended and used in the mix. The maximum cost benefit percentage of 31.38 % is observed in MX5-RCA100, in which 20 % of cement is replaced by RHA and 100 % of natural aggregate by RCA. This blend offers the cheapest option of production per cubic meter of concrete and in view of this, such materials could considerably lower the cost of production of concrete.

8. Conclusions

This research investigates the impact of replacing Portland cement with RHA on RAC's mechanical properties and microstructure. Based on the outcomes of these experiments, the following are the conclusions of this research.

- Particles of RHA have an irregular shape and more specific surface area, which increases their frictional resistance and decreases the quantity of free water in the concrete matrix. There is a direct relationship between the water consumption rate and the RA replacement ratio. The concrete's workability is reduced when RHA and RA are combined.

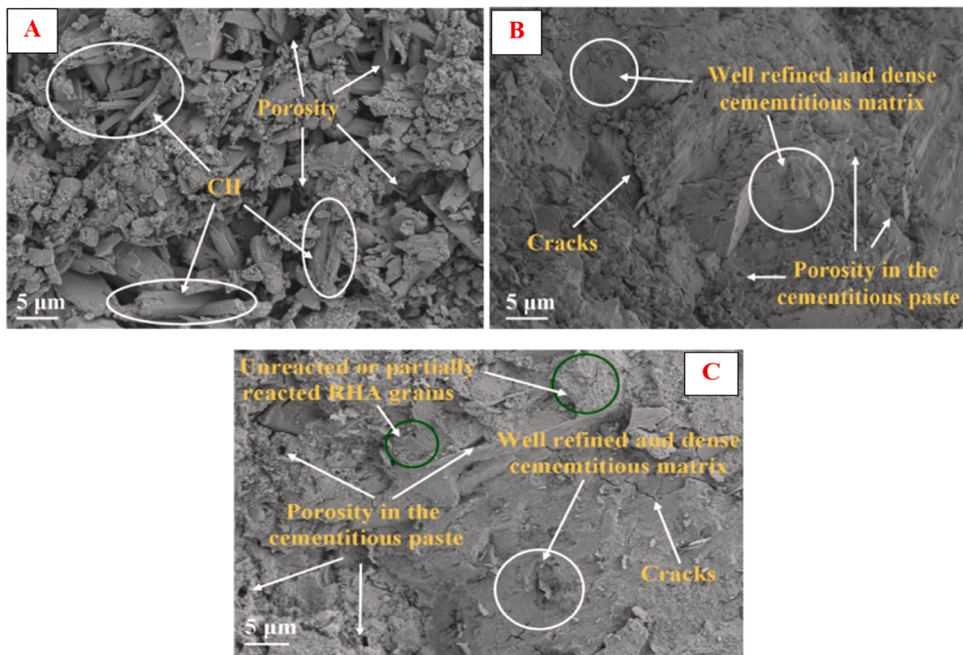


Fig. 10. SEM analysis (A) 100 % RAC, (B) 100 % RAC + 10 % RHA (C) 100 % RAC + 20 % RHA [58].

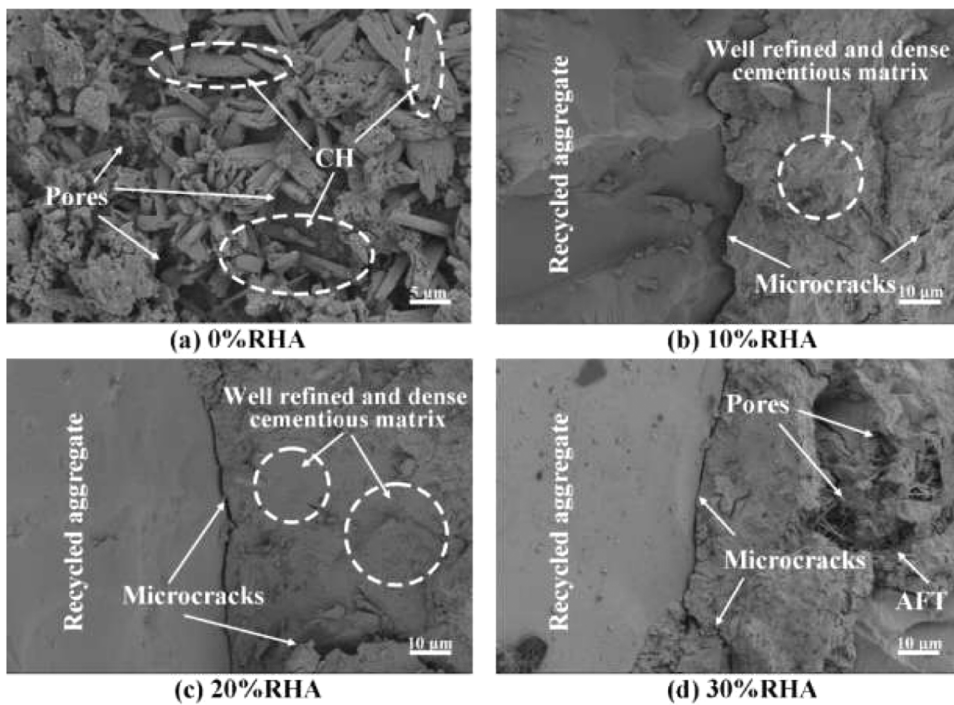


Fig. 11. SEM analysis of RCA using different percentages of RHA[106].

- The results reveal that of 15 % and 20 % RHA improves the strength of the RCA concrete by 13 % and 19 %, correspondingly after 28 days. This demonstrates a beneficial influence of 20 % RHA on concrete compressive strength. However, the effect on tensile strength is still uncertain. The capacity of RHA to occupy tiny gaps and strengthen the strength of bonds indicates the possibility of enhancing tensile strength.

Table 5Cost Analysis of production of 1 m³ RHA-based RAC.

MIX	RHA	CEMENT	NCA	RCA	RHA/COST	CEMENT/COST	RCA/NCA	TOTAL	COST BENEFIT
MX1-RCA0	0	440	1115	0	0	13200	17840	31040	
MX2-RCA0	22	418	1115	0	176	12540	17840	30556	1.56
MX3-RCA0	44	396	1115	0	352	11880	17840	30072	3.12
MX4-RCA0	66	374	1115	0	528	11220	17840	29588	4.68
MX5-RCA0	88	352	1115	0	704	10560	17840	29104	6.24
MX1-RCA100	0	440	0	1115	0	13200	10035	23235	25.14
MX2-RCA100	22	418	0	1115	176	12540	10035	22751	26.70
MX3-RCA100	44	396	0	1115	352	11880	10035	22267	28.26
MX4-RCA100	66	374	0	1115	528	11220	10035	21783	29.82
MX5-RCA100	88	352	0	1115	704	10560	10035	21299	31.38

- RCA exhibits greater initial water absorption because of its larger porosity compared to regular concrete. However, the inclusion of RHA reduces WA gradually. RHA is used to fill the pores in the concrete matrix, resulting in a reduction in water absorption and an improvement in the durability of the concrete.
- Concrete with RCA has slower sound wave transmission, indicating weaknesses. RAC is strengthened by adding RHA, which fills pores and improves sound wave passage.
- The analysis of RHA-RCA mixtures reveals significant insights into their economic viability and efficiency. Mixtures containing 20 % RHA, and 80 % RCA demonstrated the highest benefit-cost ratio at 0.18, marking them the most economically efficient. This was closely followed by mixtures with 20 % RHA and 100 % RCA and 20 % RHA with 90 % RCA, which exhibited benefit-cost ratios of 0.133 and 0.129, respectively. The results underline that cost significantly influences the economics of concrete mixes, as the increase in the proportion of recycled concrete aggregate (RCA) correlates with rising costs, while the cost of rice husk ash (RHA) remains constant relative to the mix design percentage. Conversely, despite their lower cost, mixtures with 10 % RHA showed the least benefit-cost ratio at 0.035, highlighting the trade-offs between cost and efficiency.
- The study further elucidates the material cost dynamics, where RHA, derived from industrial waste, contributes negligibly to material costs aside from transportation. RCA, however, emerges as a significant cost factor. The use of waste materials to reduce production costs highlights the sustainability of this approach.

Ethical Approval

All authors approve that the research was performed under all the ethical norms.

CRedit authorship contribution statement

Ahmed Deifalla: Writing – original draft, Resources, Methodology. **Aïssa Rezzoug:** Writing – review & editing, Methodology, Conceptualization. **Khalid Alharthi:** Writing – original draft, Methodology. **Tariq Ali:** Validation, Formal analysis, Data curation. **Kennedy C. Onyelowe:** Writing – review & editing, Resources, Formal analysis, Conceptualization. **Muhammad Zeeshan Qureshi:** Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Mohammad Alharthai:** Writing – review & editing, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mohammad Alharthai reports financial support was provided Najran University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Acknowledgments

The authors are thankful to the Deanship of Graduate Studies and Scientific Research at Najran University for funding this work under the Easy Funding Program grant code (NU/EFP/SERC/13/123).

Consent to Publish

All authors consent to publishing this paper.

Data Availability

Data will be made available on request.

References

- [1] Z. Khan, U. Balunaini, S.C.-C. Materials, 2024, Environmental feasibility and implications in using recycled construction and demolition waste aggregates in road construction based on leaching and life. <https://doi.org/10.1016/j.clema.2024.100239>.
- [2] F. Karaca, A.T.- Recycling, Reforming construction waste management for circular economy in Kazakhstan: a cost-benefit analysis of upgrading construction and demolition waste, Recycling (2023), <https://doi.org/10.3390/recycling9010002>.
- [3] A. Elmagarhe, Q. Lu, M. Alharthai, M. Alamri, A.E.- Materials, 2022, Performance of porous asphalt mixtures containing recycled concrete aggregate and fly ash, <https://doi.org/10.3390/ma15186363>.
- [4] W. Sae-Long, T. Chompoorat, , 2024, Experimental and simulation analysis of RCA and para-wood ash as partial substitutes for NCA and cement in recycled aggregate concrete, <https://doi.org/10.1016/j.cscm.2024.e03716>.
- [5] H. Chen, T. Yen, K.C.-C. and concrete research, 2003, Use of building rubbles as recycled aggregates, [https://doi.org/10.1016/S0008-8846\(02\)00938-9](https://doi.org/10.1016/S0008-8846(02)00938-9).
- [6] J.M.V. Gómez-Soberón, Porosity of recycled concrete with substitution of recycled concrete aggregate: An experimental study, Cem. Concr. Res. (2002), [https://doi.org/10.1016/S0008-8846\(02\)00795-0](https://doi.org/10.1016/S0008-8846(02)00795-0).
- [7] A. Padmini, K. Ramamurthy, M.M.-C. and building, 2009, Influence of parent concrete on the properties of recycled aggregate concrete, <https://doi.org/10.1016/j.conbuildmat.2008.03.006>.
- [8] R. Purushothaman, R.R. Amirthavalli, L. Karan, Influence of treatment methods on the strength and performance characteristics of recycled aggregate concrete, J. Mater. Civ. Eng. 27 (2015), [https://doi.org/10.1061/\(ASCE\)MT.1943-5533.0001128](https://doi.org/10.1061/(ASCE)MT.1943-5533.0001128).
- [9] A. Elmagarhe, Q. Lu, M. Alamri, ... M.A.-C.S. in, 2024, Laboratory performance evaluation of porous asphalt mixture containing recycled concrete aggregate and fly ash, <https://doi.org/10.1016/j.cscm.2023.e02849>.
- [10] N. Makul, R. Fediuk, M. Amran, A.M. Zeyad, G. Murali, N. Vatin, S. Klyuev, T. Ozbakkaloglu, Y. Vasilev, Use of recycled concrete aggregates in production of green cement-based concrete composites: A review, <https://doi.org/10.3390/cryst11030232>.
- [11] C. Lima, A. Caggiano, C. Faella, , 2013, Physical properties and mechanical behaviour of concrete made with recycled aggregates and fly ash, <https://doi.org/10.1016/j.conbuildmat.2013.04.051>.
- [12] A. Katz, 2003, Properties of concrete made with recycled aggregate from partially hydrated old concrete, [https://doi.org/10.1016/S0008-8846\(02\)01033-5](https://doi.org/10.1016/S0008-8846(02)01033-5).
- [13] H. Gebremariam, S. Taye, A.T.- Heliyon, undefined 2024, Parent concrete strength effects on the quality of recycled aggregate concrete: A review, <https://doi.org/10.1016/j.heliyon.2024.e26212>.
- [14] A.; Abdo, A.; El-Zohairy, Y.; Alashker, M.A.E.; Badran, I. Alvarez, A. Abdo, A. El-Zohairy, Y. Alashker, M. Abd, E.-A. Badran, S. Ahmed, Effect of Treated/ Untreated Recycled Aggregate Concrete: Structural Behavior of RC Beams, <https://doi.org/10.3390/su16104039>.
- [15] B. Wang, L. Yan, Q. Fu, , 2021, A comprehensive review on recycled aggregate and recycled aggregate concrete, <https://doi.org/10.1016/j.resconrec.2021.105565>.
- [16] B. Ali, L. Ali Qureshi, Influence of glass fibers on mechanical and durability performance of concrete with recycled aggregates, Construc. Build. Mater. (2019), <https://doi.org/10.1016/j.conbuildmat.2019.116783>.
- [17] S. Pradhan, S. Kumar, S.V. Barai, Performance of reinforced recycled aggregate concrete beams in flexure: experimental and critical comparative analysis, Mater. Struct. Mater. Et. Constr. 51 (2018), <https://doi.org/10.1016/j.s11527-018-1185-0>.
- [18] M. Alharthai, T. Ali, M. Qureshi, , 2024, The enhancement of engineering characteristics in recycled aggregates concrete combined effect of fly ash, silica fume and PP fiber, DOI: 10.1016/j.aej.2024.03.084.
- [19] X. Miao, J. Zhu, W. Zhu, Y. Wang, , 2024, Intelligent prediction of comprehensive mechanical properties of recycled aggregate concrete with supplementary cementitious materials using hybrid, <https://doi.org/10.1016/j.cscm.2024.e03708>.
- [20] L. Peng, Y. Zhao, J. Ban, Y. Wang, P. Shen, , 2023, Enhancing the corrosion resistance of recycled aggregate concrete by incorporating waste glass powder, <https://doi.org/10.1016/j.cemconcomp.2022.104909>.
- [21] T. Mi, L. Peng, K. Yu, , 2023, Enhancement strategies for recycled brick aggregate concrete using MICP and EICP treatments, <https://doi.org/10.1016/j.job.2023.107909>.
- [22] A. Moghadam, F. Omidinasab, , 2021, Characterization of concrete containing RCA and GGBFS: Mechanical, microstructural and environmental properties, <https://doi.org/10.1016/j.conbuildmat.2021.123134>.
- [23] B.P. Lenka, R.K. Majhi, S. Singh, A.N. Nayak, Eco-friendly and cost-effective concrete utilizing high-volume blast furnace slag and demolition waste with lime, Eur. J. Environ. Civ. Eng. 26 (2022) 5351–5373, <https://doi.org/10.1080/19648189.2021.1896581>.
- [24] S. Kou, C. Poon, , 2011, Comparisons of natural and recycled aggregate concretes prepared with the addition of different mineral admixtures, <https://doi.org/10.1016/j.cemconcomp.2011.05.009>.
- [25] I. Hakeem, M. Alharthai, M. Amin, , 2023, Properties of sustainable high-strength concrete containing large quantities of industrial wastes, nanosilica and recycled aggregates, <https://doi.org/10.1016/j.jmrt.2023.05.050>.
- [26] B. Xi, N. Zhang, H. Duan, J. He, G. Song, ... H.L.-E.I, 2023, Optimization of rice husk ash concrete design towards economic and environmental assessment, <https://doi.org/10.1016/j.eiar.2023.107229>.
- [27] M. Koushkbaghi, M. Kazemi, ... H.M.-C. and B, 2019, Acid resistance and durability properties of steel fiber-reinforced concrete incorporating rice husk ash and recycled aggregate, <https://doi.org/10.1016/j.conbuildmat.2018.12.224>.
- [28] Lea, F.M. (1970). The Chemistry of Cement and Concrete. ... - Google Scholar.
- [29] F. Costa, D.R.-J. of C. Production, 2020, Reduction in CO₂ emissions during production of cement, with partial replacement of traditional raw materials by civil construction waste (CCW), <https://doi.org/10.1016/j.jclepro.2020.123302>.
- [30] I. Quispe, R. Navia, et al., Life cycle assessment of rice husk as an energy source. A peruvian case study, J. Clean. Prod. (2019), <https://doi.org/10.1016/j.jclepro.2018.10.312>.
- [31] R. Pöde, 2016, Potential applications of rice husk ash waste from rice husk biomass power plant, <https://doi.org/10.1016/j.rser.2015.09.051>.
- [32] R.L. Reano, 2020, Assessment of environmental impact and energy performance of rice husk utilization in various biohydrogen production pathways, <https://doi.org/10.1016/j.biortech.2019.122590>.
- [33] L. Hu, Z. He, , 2020, Sustainable use of rice husk ash in cement-based materials: Environmental evaluation and performance improvement, <https://doi.org/10.1016/j.jclepro.2020.121744>.
- [34] J. Prasara-A, S.H. Gheewala, 2017, Sustainable utilization of rice husk ash from power plants: A review, DOI: 10.1016/j.jclepro.2016.11.042.
- [35] E. Öztürk, C. Ince, S. Derogar, R. Ball, Factors affecting the CO₂ emissions, cost efficiency and eco-strength efficiency of concrete containing rice husk ash: a database study, Construc. Build. Mater. (2022), <https://doi.org/10.1016/j.conbuildmat.2022.126905>.
- [36] S. Endale, W. Taffese, D. Vo, et al., Rice husk ash in concrete, Sustainability (2022) (2022), <https://doi.org/10.3390/su15010137>. Mdp.com.
- [37] S. Das, A. Adediran, C. Kaze, et al., Production, characteristics, and utilization of rice husk ash in alkali activated materials: an overview of fresh and hardened state properties, Construc. Build. Mater. (2022), <https://doi.org/10.1016/j.conbuildmat.2022.128341>.
- [38] L. Prasittisopin, D. Trejo, Hydration and phase formation of blended cementitious systems incorporating chemically transformed rice husk ash, Cement Concr. Compos. (2015), <https://doi.org/10.1016/j.cemconcomp.2015.03.002>.
- [39] M. Amin, 2019, Efficiency of rice husk ash and fly ash as reactivity materials in sustainable concrete, SpringerM Amin, BA AbdelsalamSustainable Environment Research, 2019•Springer 1 (2019). <https://doi.org/10.1186/s42834-019-0035-2>.
- [40] V. Van, C. Röbber, D. Bui, et al., Rice husk ash as both pozzolanic admixture and internal curing agent in ultra-high performance concrete, Cement Concr. Compos. (2014), <https://doi.org/10.1016/j.cemconcomp.2014.07.015>.
- [41] K. Ganesan, K. Rajagopal, K.T.-C. and building, 2008, Rice husk ash blended cement: Assessment of optimal level of replacement for strength and permeability properties of concrete, <https://doi.org/10.1016/j.conbuildmat.2007.06.011>.

- [42] B. Chatveera, P.L.-J. of Environmental Management, 2011, Durability of conventional concretes containing black rice husk ash, <https://doi.org/10.1016/j.jenvman.2010.08.007>.
- [43] C. Gomes, P. Salles, et al., Mechanical and durability properties of concrete produced with construction and demolition waste and rice husk ash, *Construc. Build. Mater.* (2023), <https://doi.org/10.1016/j.conbuildmat.2023.133471>.
- [44] B. Tayeh, R. Alyousef, ... H.A.-J. of C, 2021, Recycling of rice husk waste for a sustainable concrete: a critical review, <https://doi.org/10.1016/j.jclepro.2021.127734>.
- [45] H. Huang, X. Gao, H. Wang, H.Y.-C. and B. Materials, 2017, Influence of rice husk ash on strength and permeability of ultra-high-performance concrete, <https://doi.org/10.1016/j.conbuildmat.2017.05.155>.
- [46] F. Muleya, N. Muwila, C.K. Tembo, A. Lungu, Partial replacement of cement with rice husk ash in concrete production: an exploratory cost-benefit analysis for low-income communities, *Eng. Manag. Prod. Serv.* 13 (2021) 127–141, <https://doi.org/10.2478/emj-2021-0026>.
- [47] N.Md.S. Hasan, Md.H.R. Sobuz, Md.M.H. Khan, N.J. Mim, Md.M. Meraz, S.D. Datta, Md.J. Rana, A. Saha, A.S.M. Akid, Md.T. Mehedi, M. Houida, N.M. Sutan, Integration of rice husk ash as supplementary cementitious material in the production of sustainable high-strength concrete, *Materials* 15 (2022) 8171, <https://doi.org/10.3390/ma15228171>.
- [48] S.C. Bostanci, M. Limbachiya, H. Kew, Use of recycled aggregates for low carbon and cost effective concrete construction, *J. Clean. Prod.* 189 (2018) 176–196, <https://doi.org/10.1016/j.jclepro.2018.04.090>.
- [49] B. Wang, L. Yan, Q. Fu, B. Kasal, A comprehensive review on recycled aggregate and recycled aggregate concrete, *Resour. Conserv Recycl* 171 (2021) 105565, <https://doi.org/10.1016/j.resconrec.2021.105565>.
- [50] F. Ding, X. Fan, Y. Xie, S. Jiang, C. Qiu, ... D.S.-J. of B, 2024, Combined effect of rice husk ash and cementitious capillary crystalline waterproofing materials on the performance of mortar, <https://doi.org/10.1016/j.job.2024.108479>.
- [51] W. Ma, Y. Wang, L. Huang, L. Yan, B.K.-J. of B. Engineering, 2023, Natural and recycled aggregate concrete containing rice husk ash as replacement of cement: Mechanical properties, microstructure, strength model and statistical, <https://doi.org/10.1016/j.job.2023.105917>.
- [52] M. Alnahhal, A. Hamdan, ... A.H.-C. and C, 2021, Effect of rice husk ash-derived activator on the structural build-up of alkali activated materials, <https://doi.org/10.1016/j.cemconres.2021.106590>.
- [53] C. Liu, W. Zhang, H. Liu, C. Zhu, Y. Wu, ... C.H.-C. and B, 2022, Recycled aggregate concrete with the incorporation of rice husk ash: Mechanical properties and microstructure, <https://doi.org/10.1016/j.conbuildmat.2022.128934>.
- [54] ASTM C150/C150M, Standard specification for Portland cement, 2019.
- [55] ASTM C 128, Standard test method for density, relative density (specific gravity) and absorption of Fine Aggregate, 1992.
- [56] ASTM C127-81, Standard test method for specific gravity and absorption of coarse aggregate, 1983.
- [57] ASTM C566, Standard Test Method for Total Evaporable Moisture Content of Aggregate by Drying, West Conshohocken, PA, 2013.
- [58] C. Liu, W. Zhang, H. Liu, C. Zhu, Y. Wu, ... C.H.-C. and B, 2022, Recycled aggregate concrete with the incorporation of rice husk ash: Mechanical properties and microstructure, <https://doi.org/10.1016/j.conbuildmat.2022.128934>.
- [59] R. Zerbino, G. Giaccio, G.I.-C. and building materials, 2011, Concrete incorporating rice-husk ash without processing, <https://doi.org/10.1016/j.conbuildmat.2010.06.016>.
- [60] R. Padhi, R. Patra, B. Mukharjee, T.D.-C. and Building, 2018, Influence of incorporation of rice husk ash and coarse recycled concrete aggregates on properties of concrete, <https://doi.org/10.1016/j.conbuildmat.2018.03.270>.
- [61] C. Poon, Z. Shui, L. Lam, H. Fok, S.K.-C. and concrete research, 2004, Influence of moisture states of natural and recycled aggregates on the slump and compressive strength of concrete, [https://doi.org/10.1016/S0008-8846\(03\)00186-8](https://doi.org/10.1016/S0008-8846(03)00186-8).
- [62] T. Hansen, H. Narud, *Strength of Recycled Concrete Made from Crushed Concrete Coarse Aggregate*, *Concrete International*, 1983.
- [63] A. Ajwad, S. Ahmad, Abdullah, S.T.A. Jaffar, U. Ilyas, M.A. Adnan, Effect of using rice husk ash as partial replacement of cement on properties of fresh and hardened concrete, *Archit. Civ. Environ.* 15 (2022) 85–94, <https://doi.org/10.2478/ACEE-2022-0017>.
- [64] M. Jamil, M. Khan, M. Karim, et al., Physical and chemical contributions of Rice Husk Ash on the properties of mortar, *Construc. Build. Mater.* (2016), <https://doi.org/10.1016/j.conbuildmat.2016.10.029>.
- [65] N. Van Tuan, G. Ye, ... K.V.B.-C. and C, 2011, Hydration and microstructure of ultra high performance concrete incorporating rice husk ash, <https://doi.org/10.1016/j.cemconres.2011.06.009>.
- [66] J. Alex, J. Dhanalakshmi, et al., Experimental investigation on rice husk ash as cement replacement on concrete production, *Construc. Build. Mater.* (2016), <https://doi.org/10.1016/j.conbuildmat.2016.09.150>.
- [67] H. Venkatanarayanan, P.R.-C. and concrete composites, 2015, Effect of grinding of low-carbon rice husk ash on the microstructure and performance properties of blended cement concrete, <https://doi.org/10.1016/j.cemconcomp.2014.09.021>.
- [68] V. Sata, J. Tangpagasit, ... C.J.-C. and C, 2012, Effect of W/B ratios on pozzolanic reaction of biomass ashes in Portland cement matrix, <https://doi.org/10.1016/j.cemconcomp.2011.09.003>.
- [69] V. Sata, J. Tangpagasit, ... C.J.-C. and C, 2012, Effect of W/B ratios on pozzolanic reaction of biomass ashes in Portland cement matrix, <https://doi.org/10.1016/j.cemconcomp.2011.09.003>.
- [70] H. Chao-Lung, B. Le Anh-Tuan, C.C.-T.-C. and building, 2011, Effect of rice husk ash on the strength and durability characteristics of concrete, <https://doi.org/10.1016/j.conbuildmat.2011.04.009>.
- [71] V. Kannan, Strength and durability performance of self-compacting concrete containing self-combusted rice husk ash and metakaolin, *Construc. Build. Mater.* (2018), <https://doi.org/10.1016/j.conbuildmat.2017.11.043>.
- [72] Influence of rice husk-bark ash on mechanical properties of concrete containing high amount of recycled aggregates, <https://doi.org/10.1016/j.conbuildmat.2007.05.004>.
- [73] K. Sagoe-Crentsil, T. Brown, A.T.-C. and concrete research 2001, Performance of concrete made with commercially produced coarse recycled concrete aggregate, [https://doi.org/10.1016/S0008-8846\(00\)00476-2](https://doi.org/10.1016/S0008-8846(00)00476-2).
- [74] K. Yan, H. Lan, Q. Li, D. Ge, Y.L.-C. and B. Materials, 2022, Optimum utilization of recycled aggregate and rice husk ash stabilized base material, <https://doi.org/10.1016/j.conbuildmat.2022.128627>.
- [75] H. Ahmed, J.A. Bogas, M. Guedes, Mechanical behavior and transport properties of cementitious composites reinforced with carbon nanotubes, *J. Mater. Civ. Eng.* 30 (2018), [https://doi.org/10.1061/\(ASCE\)MT.1943-5533.0002470](https://doi.org/10.1061/(ASCE)MT.1943-5533.0002470).
- [76] A. Carriço, J. Bogas, A. Hawreen, M.G.-C. and Building, 2018, Durability of multi-walled carbon nanotube reinforced concrete, <https://doi.org/10.1016/j.conbuildmat.2017.12.221>.
- [77] D. Bui, J. Hu, P.S.-C. and concrete composites, 2005, Particle size effect on the strength of rice husk ash blended gap-graded Portland cement concrete, <https://doi.org/10.1016/j.cemconcomp.2004.05.002>.
- [78] S.P. Muñoz-Pérez, E. Sánchez-Díaz, D. Barboza-Culqui, J.M. Garcia-Chumacero, Use of recycled concrete and rice husk ash for concrete: A review, DOI: 10.22201/icat.24486736e.2024.22.1.2248.
- [79] A.M. Neville, *Properties of Concrete*, 5th, Google Scholar, 2012.
- [80] P.K. Mehta, P.J.M. Monteiro, *Concrete*, Google Scholar, 2014.
- [81] L. Hu, Z. He, S.Z.-J. of C. Production, 2020, Sustainable use of rice husk ash in cement-based materials: Environmental evaluation and performance improvement, <https://doi.org/10.1016/j.jclepro.2020.121744>.
- [82] S. Sair, A. Oushabi, O. Tanane, Y. Abboud, S. Sair, A. Oushabi, A. Kammouni, O. Tanane, Y. Abboud, A. El Bouari, Case studies in construction materials, Researchgate.Net (2018), <https://doi.org/10.1016/j.cscm.2018.02.001>.
- [83] Z. Zhao, L.F.-C.S. in C. Materials, 2018, The probability distribution of pitting for accelerated corrosion reinforcement, <https://doi.org/10.1016/j.cscm.2018.e00193>.

- [84] S. Abid, A. Nahhab, ... H.A.-C.S. in, 2018, Expansion and strength properties of concrete containing contaminated recycled concrete aggregate, <https://doi.org/10.1016/j.cscm.2018.e00201>.
- [85] R. Faraj, H. Ahmed, A.S.-H. of sustainable, 2022, Fresh and mechanical properties of concrete made with recycled plastic aggregates, DOI: 10.1016/B978-0-12-821730-6.00023-1.
- [86] D. Bui, S.A. Kodjo, P. Rivard, B. Fournier, Evaluation of concrete distributed cracks by ultrasonic travel time shift under an external mechanical perturbation: study of indirect and semi-direct transmission configurations, *J. Nondestruct. Eval.* 32 (2013) 25–36, <https://doi.org/10.1007/S10921-012-0155-7>.
- [87] R. Madandoust, M. Ranjbar, ... H.M.-B, 2011, Mechanical properties and durability assessment of rice husk ash concrete, <https://doi.org/10.1016/j.biosystemseng.2011.07.009>.
- [88] R. Sri Ravindrajah, Y.H. Loo, C.T. Tam, Strength evaluation of recycled-aggregate concrete by in-situ tests, *Mater. Struct.* 21 (1988) 289–295, <https://doi.org/10.1007/BF02481828>.
- [89] M. Chakradhara Rao, S.K. Bhattacharyya, S.V. Barai, Influence of field recycled coarse aggregate on properties of concrete, *Mater. Struct. /Mater. Et. Constr.* 44 (2011) 205–220, <https://doi.org/10.1617/S11527-010-9620-X>.
- [90] M. Kazemi, R. Madandoust, et al., Compressive strength assessment of recycled aggregate concrete using Schmidt rebound hammer and core testing, *Construc. Build. Mater.* (2019), <https://doi.org/10.1016/j.conbuildmat.2019.07.110>.
- [91] L. Evangelista, J. de Brito, Flexural behaviour of reinforced concrete beams made with fine recycled concrete aggregates, *KSCE J. Civ. Eng.* 21 (2017) 353–363, <https://doi.org/10.1007/S12205-016-0653-8>.
- [92] D. Pedro, J. De Brito, et al., Structural concrete with simultaneous incorporation of fine and coarse recycled concrete aggregates: mechanical, durability and long-term properties, *Construc. Build. Mater.* (2017), <https://doi.org/10.1016/j.conbuildmat.2017.07.215>.
- [93] M. Meddah, A. Al-Harthy, M.A.L.- Buildings, 2020, Recycled concrete aggregates and their influences on performances of low and normal strength concretes, <https://doi.org/10.3390/buildings10090167>.
- [94] K. McNeil, T.K.-I. journal of concrete structures and, undefined, Recycled concrete aggregates: A review, *Springer* 7 (2013) (2013) 61–69, <https://doi.org/10.1007/s40069-013-0032-5>.
- [95] M. Safiuddin, U.J. Alengaram, M.M. Rahman, M.A. Salam, M.Z. Jumaat, Use of recycled concrete aggregate in concrete: a review, *J. Civ. Eng. Manag.* 19 (2013) 796–810, <https://doi.org/10.3846/13923730.2013.799093>.
- [96] V. Saraswathy, H.S.-C. and building materials, Corrosion performance of rice husk ash blended concrete, ElsevierV Saraswathy. HW SongConstruction and Building Materials, Elsevier, 2007, <https://doi.org/10.1016/j.conbuildmat.2006.05.037>.
- [97] N. Van Tuan, G. Ye, K. Van Breugel, A.F.-... and B. Materials, 2011, The study of using rice husk ash to produce ultra-high performance concrete, <https://doi.org/10.1016/j.conbuildmat.2010.11.046>.
- [98] A. Siddika, M.Al Mamun, ... R.A.-J. of K.S, 2021, State-of-the-art-review on rice husk ash: A supplementary cementitious material in concrete, <https://doi.org/10.1016/j.jksues.2020.10.006>.
- [99] T. Fode, Y. Jande, T.K.- Heliyon, 2023, Effects of different supplementary cementitious materials on durability and mechanical properties of cement composite–Comprehensive review, <https://doi.org/10.1016/j.heliyon.2023.e17924>.
- [100] D. Chopra, R.S.-B. engineering, 2015, Strength, permeability and microstructure of self-compacting concrete containing rice husk ash, <https://doi.org/10.1016/j.biosystemseng.2014.12.005>.
- [101] B.S. Thomas, Green concrete partially comprised of rice husk ash as a supplementary cementitious material–A comprehensive review, *Renew. Sustain. Energy Rev.* (2018), <https://doi.org/10.1016/j.rser.2017.10.081>.
- [102] B. Zhan, D. Xuan, C. Poon, K.S.-C. and Concrete, 2020, Characterization of interfacial transition zone in concrete prepared with carbonated modeled recycled concrete aggregates, <https://doi.org/10.1016/j.cemconres.2020.106175>.
- [103] B.M. Miyandehi, A. Feizbakhsh, M.A. Yazdi, Q.-F. Liu, J. Yang, P. Alipour, Performance and properties of mortar mixed with nano-CuO and rice husk ash, 2016, Elsevier, 2016, <https://doi.org/10.1016/j.cemconcomp.2016.10.006>.
- [104] D. Wang, X. Zhou, B. Fu, et al., Chloride ion penetration resistance of concrete containing fly ash and silica fume against combined freezing-thawing and chloride attack, *Construc. Build. Mater.* (2018), <https://doi.org/10.1016/j.conbuildmat.2018.03.038>. Elsevier.
- [105] O. Gencel, A. Benli, O. Bayraktar, G.K.-... and B. Materials, 2021, Effect of waste marble powder and rice husk ash on the microstructural, physico-mechanical and transport properties of foam concretes exposed to high temperatures, <https://doi.org/10.1016/j.conbuildmat.2021.123374>.
- [106] W. Zhang, H. Liu, C.L.-A. Sciences, 2022, Impact of rice husk ash on the mechanical characteristics and freeze–thaw resistance of recycled aggregate concrete, <https://doi.org/10.3390/app122312238>.